Business Rule ID	Business Rule
CALCDTCONFLICT-1	The conflict begin interval date for a previous vaccine dose administered must be calculated as the date administered of the previous vaccine dose administered plus the conflict begin interval of a vaccine type conflict where all the following are true: • The vaccine type of the current vaccine dose administered is an impacted vaccine type. • The vaccine type of the previous vaccine dose administered is a conflicting vaccine type for the impacted vaccine type.

CALCDTCONFLICT-2	The conflict end interval date for a previous vaccine dose administered must be calculated as one of the following:
	
	• the date administered of the previous vaccine dose administered plus the minimum conflict end interval of a vaccine type conflict if all the following are true:
	  ○ The vaccine type of the current vaccine dose administered is an impacted vaccine type.
	  ○ The vaccine type of the previous vaccine dose administered is a conflicting vaccine type for the impacted vaccine type.
	  ○ The previous vaccine dose administered has one of the following:
	  ▪ an evaluation status of 'Valid'
	  ▪ no evaluation status.
	• the date administered of the previous vaccine dose administered plus the conflict end interval of a vaccine type conflict if all the following are true:
	  ○ The vaccine type of the current vaccine dose administered is an impacted vaccine type.
	  ○ The vaccine type of the previous vaccine dose administered is a conflicting vaccine type for the impacted vaccine type.
	  ○ The previous vaccine dose administered has an evaluation status.
	  ○ The previous vaccine dose administered does not have an evaluation status of 'Valid'.

CONFLICT-3	A current vaccine dose administered must be considered an impacted vaccine dose administered if all the following are true for the date administered of the current vaccine dose administered: • It is on or after the conflict begin interval date. • It is before the conflict end interval date.

6.8 Evaluate for Preferable Vaccine

Evaluate for preferable vaccine validates the vaccine of a vaccine dose administered against the list of preferable vaccines.

Figure 6-17 depicts a patient who received a preferable vaccine while Figure 6-18 depicts a patient who did not receive a preferable vaccine.

Current Vaccine Dose Administered			
Vaccine Type	Administration Date	Trade Name	Volume
456	06/30/2009	DEF	0.5 ml

Patient received a Preferable Vaccine

Preferable Vaccines for the Target Dose				
Vaccine Type	Vaccine Type Begin Age Date	Vaccine Type End Age Date	Trade Name	Volume
123	01/01/2000	12/31/2999	ABC	0.5 ml
456	01/01/2000	12/31/2050	DEF	0.5 ml
789	01/01/2000	12/31/2008	GHI	0.5 ml

FIGURE 6-17 PATIENT RECEIVED A PREFERABLE VACCINE

Current Vaccine Dose Administered			
Vaccine Type	Administration Date	Trade Name	Volume
567	09/04/2006	XYZ	0.5 ml

Patient did not receive a preferable vaccine

Preferable Vaccines for the Target Dose				
Vaccine Type	Vaccine Type Begin Age Date	Vaccine Type End Age Date	Trade Name	Volume
X 123	01/01/2000	12/31/2999	ABC	0.5 ml
X 456	01/01/2000	12/31/2050	DEF	0.5 ml
X 789	01/01/2000	12/31/2008	GHI	0.5 ml

FIGURE 6-18 PATIENT DID NOT RECEIVE A PREFERABLE VACCINE

It should be noted that volume is sparsely populated and tracked differently in most systems. Therefore, volume will not be used to evaluate the validity of a vaccine dose administered. However, it will be provided as an evaluation reason that less than sufficient volume was administered.

The following process model, attribute table, decision table, and business rule table are used to evaluate for a preferable vaccine.

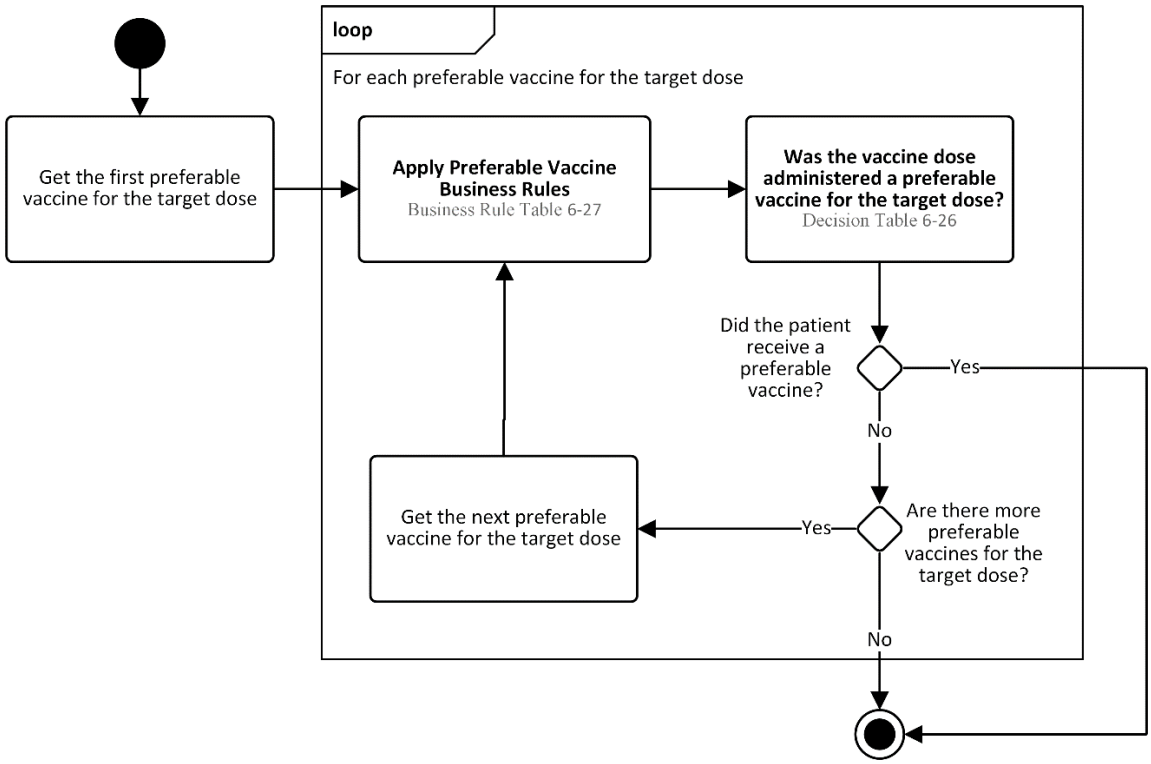

FIGURE 6-19 EVALUATE FOR A PREFERABLE VACCINE PROCESS MODEL

TABLE 6-25 PREFERABLE VACCINE ADMINISTERED ATTRIBUTES

Attribute Type	Attribute Name	Assumed Value if Empty
Vaccine dose administered	Date Administered	-
Vaccine dose administered	Volume	-
Vaccine dose administered	Trade Name	-
Supporting Data	Preferable Vaccine elements	-
Calculated date (CALCDTPREF-1)	Preferable Vaccine Type Begin Age Date	01/01/1900
Calculated date (CALCDTPREF-2)	Preferable Vaccine Type End Age Date	12/31/2999

TABLE 6-26 WAS THE VACCINE DOSE ADMINISTERED A PREFERABLE VACCINE FOR THE TARGET DOSE?

CONDITIONS	RULES				
Is the vaccine type of the vaccine dose administered the same as the vaccine type of a preferable vaccine for the target dose?	Yes	Yes	No	Yes	Yes

CONDITIONS	RULES				
Is the preferable vaccine type begin age date ≤ date administered < preferable vaccine type end age date?	Yes	Yes	-	No	Yes
Is the trade name of the vaccine dose administered the same as the trade name of the preferable vaccine for the target dose?	Yes	Yes	-	-	No
Is the volume of the vaccine dose administered ≥ the volume of the preferable vaccine for the target dose?	Yes	No	-	-	-
OUTCOMES	Yes. The vaccine dose administered was a preferable vaccine for the target dose.	Yes. The vaccine dose administered was a preferable vaccine for the target dose. Evaluation Reason is 'Volume administered is less than recommended volume'.	No. The vaccine dose administered was not a preferable vaccine for the target dose.	No. The vaccine dose administered was not a preferable vaccine for the target dose. It was administered out of the recommended age range for the preferable vaccine.	No. The vaccine dose administered was not a preferable vaccine for the target dose. The trade name of the vaccine dose administered is not the same as the trade name of the preferable vaccine.

TABLE 6-27 PREFERABLE VACCINE BUSINESS RULES

Business Rule ID	Business Rule
CALCDTPREF-1	A patient's preferable vaccine type begin age date must be calculated as the patient's date of birth plus the vaccine type begin age of a preferable vaccine.
CALCDTPREF-2	A patient's preferable vaccine type end age date must be calculated as the patient's date of birth plus the vaccine type end age of a preferable vaccine.

6.9 Evaluate for Allowable Vaccine

Evaluate for allowable vaccine validates the vaccine of a vaccine dose administered against the list of allowable vaccines.

Figure 6-20 depicts a patient who received an allowable vaccine while Figure 6-21 depicts a patient who did not receive an allowable vaccine.